

Shifted-zeta evolution and Wilson lines

A Loewner realization program for cumulative positivity

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Abstract

A specified Loewner evolution generates a curve, and a fixed quantum field theory assigns Wilson observables to that curve. We investigate the relation between geometric growth and expectation-value evolution without requiring a closed-form path integral. For a genuine growing trace in four-dimensional Yang–Mills theory, completed by a straight return chord, the moving-tip connection terms cancel. We derive the first two curvature-insertion hierarchy equations and, for a linear driver, a short-time response with an explicit smooth-field remainder bound. The expectation statements use a fixed positive field-flow resolution and stated regularity and moment assumptions; a closed Makeenko–Migdal reduction remains open. The longer-term realization question asks whether a boundary observable can reproduce the exact shifted-zeta evolution $\partial_\omega V_{\omega,L} = -G_{\omega,L} V_{\omega,L}$ and admit a positive-norm identity for its cumulative defect. Such contraction along $L \rightarrow \infty$, $\omega \downarrow 0$ would imply Weil positivity. Earlier Gaussian readout and free-localization calculations constrain particular candidates; they do not exhaust the observables of a fixed interacting theory. The companion gives the derivations, quantum assumptions and separate arithmetic context. No matching physical transfer, cumulative norm identity or all-depth continuation is established.

Status. Exposition of an open research program, with background identities and conditional implications distinguished from construction hypotheses. Prepared with OpenAI GPT-6 (Codex) assistance for Edward Baker; see Appendix A.

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1 The program: geometric growth and quantum evolution

Fix a quantum field theory and let a specified Loewner driver generate a curve. A Wilson observable on that curve then defines a family of quantum expectations. The immediate question is how its evolution is related to the geometric growth law. One can derive such an equation without evaluating the expectation in closed form. The gauge representation, endpoint contraction, operator insertions and, in a supersymmetric theory, scalar-coupling profile give distinct observables within the same bulk theory.

We begin that calculation in pure four-dimensional Yang–Mills theory. The actual growing trace is closed by a straight return chord. At fixed smooth-field resolution, moving-tip connection terms cancel, leaving an exact curvature-insertion equation. A second equation exposes the covariant-derivative and ordered two-curvature moments needed for closure. For a linear Loewner driver, a short-time response follows with an explicit remainder bound. Quantum averaging keeps the full interacting measure under stated assumptions. Neither Gaussianity nor large- N factorization is imposed.

This physical question also has a longer-term arithmetic motivation. The shifted completed-zeta quotient supplies a prescribed operator evolution and a cumulative-positivity target. Identifying an appropriate Wilson boundary transfer with that operator is a later realization problem. A scalar loop expectation is not already an operator on all boundary inputs. The present physical calculation is therefore useful before any arithmetic match is known, while the exact matching and positive-norm requirements remain explicit below.

1.1 The proposed chain of implications

Write $V_{\omega,L}$ for the causal arithmetic transfer at shift ω on an interval of length L . Its exact evolution and initial condition are

$$\partial_{\omega} V_{\omega,L} = -G_{\omega,L} V_{\omega,L}, \quad V_{0,L} = I.$$

The generator contains the gamma contribution, a fixed local term, two pole terms and the prime-power translations. The objective is an independently defined Wilson transfer $\mathcal{V}_{\omega,L}$ satisfying this equation, together with uniqueness in the relevant causal class, so that $\mathcal{V}_{\omega,L} = V_{\omega,L}$.

A second identity should explain its norm:

$$\|f\|^2 = \|\mathcal{V}_{\omega,L} f\|^2 + \|\mathcal{B}_{\omega,L} f\|^2.$$

Here $\mathcal{B}$ would be an independently constructed amplitude map into a positive Hilbert space, potentially supplied by a reflected Wilson network. This would prove contraction at the parameters where the construction is valid. If those parameters include $L_j \rightarrow \infty$ and $\omega_j \downarrow 0$, the zero-shift limit yields the full central Weil form on every compact support. The precise conditional implication is Proposition 9.1.

The later realization therefore has three substantive tasks: derive the prescribed evolution from the observable, establish the cumulative norm identity, and continue the construction to unbounded interval length while the shift tends to zero. The known arithmetic differential equation supplies a target for these tasks; it does not discharge any of them.

1.2 The role of the two papers

This exposition retains the facts needed to assess that chain of implications. Section 2 states the full positivity target, including the local constant w_0 . Section 3 derives the transfer and defect evolutions. Sections 4–8 formulate the physical hypothesis and explain the constraints from the existing calculations. Section 5 derives the new genuine-trace quantum hierarchy in a fixed interacting theory. Section 6 gives the earlier fixed-window obstruction and the localization controls, including their spatial limitations. Section 9 states the depth–shift continuation problem and the separate fixed-window arithmetic progress, and Section 10 gives the next discriminating tasks.

The companion *Supplementary information* [1] contains the detailed constructions, proofs and diagnostics. A result that materially restricts the proposed realization is summarized here even when its derivation is deferred there. This division lets the main argument be read without the

Clifford algebra or perturbative integrals, while keeping those calculations available as explicit tests of a candidate model.

1.3 Sources and relation to earlier work

The scalar-coupled Wilson framework originates in Maldacena’s construction [2]; tangent-dependent scalar couplings and common line supersymmetry are due to Zarembo [3]. The D3–D5 defect theory and its projector conventions come from DeWolfe–Freedman–Ooguri and Gaiotto–Witten [4, 5]. Baker’s open Wilson lines and semicircle are the endpoint controls [6]. We use chordal Loewner normalization as in Lawler [7]. The Loewner/CFT relation of Bauer–Bernard [8] motivates looking for a closed averaged observable, but supplies no Wilson realization of the arithmetic generator here.

Ordinary reflection and its distinction from an internal rotation are guided by Osterwalder–Schrader and Dorn–Pershin [9, 10]. The compact-support Weil criterion is classical; our convention agrees with Suzuki [11]. The exact shift evolution is reproduced from the shared background [16]. The Wilson-lines, Loewner and fractional-dimension investigations supply the previous realization tests [12, 14, 15]. The cumulative-storage continuation comes from the shifted-zeta program [19, 18]. The arithmetic realization formulation follows Baker’s shift-flow proposal [17]. The present physical emphasis follows his clarification to study genuine Loewner growth and quantum evolution within a fixed theory, using loop equations before seeking an explicit path-integral value. The new derivation is recorded in [30]; no solution of the inverse problem is claimed.

2 The arithmetic positivity target

This section fixes the target independently of any field-theory model. The formulas also explain why positivity of a single covariance kernel would be insufficient.

2.1 The compactly supported Weil form

Let $I_L = (-L/2, L/2)$, with $L > 0$, and let $f \in C_c^\infty(I_L; \mathbb{C})$. Write F for its zero extension to $\mathbb{R}$ and use

$$\widehat{F}(\tau) = \int_{\mathbb{R}} e^{-i\tau x} F(x) dx, \quad \langle f, g \rangle = \int_{I_L} \overline{f(x)} g(x) dx. \quad (2.1)$$

For $a > 0$, define the compressed shift $T_a f(x) = F(x - a)$ on I_L . Let $\psi = \Gamma'/\Gamma$ be the digamma function, and let $\Lambda(n) = \log p$ when $n = p^k$, p prime and $k \geq 1$, and zero otherwise. Finally put

$$C(f) = \int_{I_L} f(x) \cosh(x/2) dx, \quad S(f) = \int_{I_L} f(x) \sinh(x/2) dx. \quad (2.2)$$

The normalization of the program is

$$\begin{aligned} Q_{0,L}[f] &= \frac{1}{2\pi} \int_{\mathbb{R}} [\operatorname{Re} \psi(\tfrac{1}{4} + \tfrac{i\tau}{2}) - \log \pi] |\widehat{F}(\tau)|^2 d\tau + 2|C(f)|^2 - 2|S(f)|^2 \\ &\quad - \sum_{\substack{n \geq 2 \\ \log n < L}} \frac{\Lambda(n)}{\sqrt{n}} \langle f, (T_{\log n} + T_{\log n}^*) f \rangle. \end{aligned} \quad (2.3)$$

The prime-power sum is finite on a fixed interval. The pole term is indefinite, and the archimedean multiplier includes its local constant. Neither may be dropped in a positivity argument.

For comparison with the usual explicit formula, set $h = F * \widetilde{F}$, where $\widetilde{F}(x) = \overline{F(-x)}$. The two prime sums in the Weil functional give the last line of (2.3); its $e^{x/2} + e^{-x/2}$ term gives

$2|C|^2 - 2|S|^2$. Its archimedean distribution has Fourier multiplier $\operatorname{Re} \psi(1/4 + i\tau/2) - \log \pi$. Thus $Q_{0,L}[f] = W(F * \tilde{F})$ in the convention of [11, p. 1]. In particular the weight is $\Lambda(n)/\sqrt{n}$.

Theorem 2.1 (Weil positivity criterion, used as background). *RH holds if and only if $Q_{0,L}[f] \geq 0$ for every $L > 0$ and every $f \in C_c^\infty(I_L; \mathbb{C})$.*

The equivalence is the classical criterion, not a result of the present construction. Every compact support lies in one such interval, so an unbounded sequence of interval lengths suffices. A finite collection of lengths or numerical matrices does not.

2.2 The positive kinetic term and the indispensable local term

The numbers $a_n = 2n + 1/2$, $n \geq 0$, are the decay rates $1/2, 5/2, 9/2, \dots$ in the exponential expansion

$$n_\Gamma(u) = \frac{e^{-u/2}}{1 - e^{-2u}} = \sum_{n=0}^{\infty} e^{-a_n u}, \quad u > 0. \quad (2.4)$$

They arise by expanding a geometric series, and are not zeta zeros or prime-power coefficients. In the free defect construction they also appear as the even angular rates $\ell + 1/2$ with $\ell = 2n$; the physical derivation is in [1, Section S2]. Put

$$b(\tau^2) = \operatorname{Re} \psi\left(\frac{1}{4} + \frac{i\tau}{2}\right) - \psi\left(\frac{1}{4}\right) = 2 \sum_{n=0}^{\infty} \frac{\tau^2}{a_n(a_n^2 + \tau^2)} \geq 0, \quad w_0 = \psi\left(\frac{1}{4}\right) - \log \pi < 0. \quad (2.5)$$

The exact digamma value is

$$\psi\left(\frac{1}{4}\right) = -\gamma - \frac{\pi}{2} - 3 \log 2, \quad w_0 = -\gamma - \frac{\pi}{2} - 3 \log 2 - \log \pi \simeq -5.37218342, \quad (2.6)$$

where γ is Euler's constant. Thus $\psi(1/4)$ is not equal to $\log \pi$. Adding and subtracting $\psi(1/4)$ in the original multiplier gives the exact identity

$$\begin{aligned} \operatorname{Re} \psi\left(\frac{1}{4} + \frac{i\tau}{2}\right) - \log \pi &= [\operatorname{Re} \psi\left(\frac{1}{4} + \frac{i\tau}{2}\right) - \psi\left(\frac{1}{4}\right)] + [\psi\left(\frac{1}{4}\right) - \log \pi] \\ &= b(\tau^2) + w_0. \end{aligned} \quad (2.7)$$

Thus w_0 is the remainder after separating the nonnegative multiplier $b(\tau^2)$; it is already present in (2.3), not an additional term introduced into the form.

The series in (2.5) follows by subtracting the convergent digamma partial fractions. Each exponential in (2.4) satisfies $\int_0^\infty e^{-au}(1 - \cos \tau u) du = \tau^2/[a(a^2 + \tau^2)]$. Plancherel therefore gives

$$\mathcal{E}_\Gamma[F] := \int_0^\infty n_\Gamma(u) \|F(\cdot + u) - F\|_2^2 du = \frac{1}{2\pi} \int_{\mathbb{R}} b(\tau^2) |\hat{F}(\tau)|^2 d\tau. \quad (2.8)$$

Using (2.7) and $\frac{1}{2\pi} \int_{\mathbb{R}} |\hat{F}(\tau)|^2 d\tau = \|f\|_2^2$, the archimedean contribution becomes

$$\frac{1}{2\pi} \int_{\mathbb{R}} [\operatorname{Re} \psi\left(\frac{1}{4} + \frac{i\tau}{2}\right) - \log \pi] |\hat{F}(\tau)|^2 d\tau = \mathcal{E}_\Gamma[F] + w_0 \|f\|_2^2. \quad (2.9)$$

Substituting this identity into (2.3) rewrites the full target as

$$\begin{aligned} Q_{0,L}[f] &= \mathcal{E}_\Gamma[F] + w_0 \|f\|_2^2 + 2|C(f)|^2 - 2|S(f)|^2 \\ &\quad - \sum_{\substack{n \geq 2 \\ \log n < L}} \frac{\Lambda(n)}{\sqrt{n}} \langle f, (T_{\log n} + T_{\log n}^*) f \rangle. \end{aligned} \quad (2.10)$$

Here $w_0 \|f\|_2^2 = w_0 \int_{I_L} |f(x)|^2 dx$ is the fixed negative local term. The positive difference energy is only one part of the desired form: its local term, pole contribution and prime-power translations must all be retained.

The natural form domain on a fixed interval is given by zero extensions with $\int \log(2 + |\tau|) |\widehat{F}(\tau)|^2 d\tau < \infty$. Indeed $b(\tau^2)$ grows logarithmically at infinity, whereas the pole and finite translation terms are bounded forms on $L^2(I_L)$. This explains the logarithmic form domain and semiboundedness of the localized form. For the logical proof target in Theorem 2.1, the smooth compactly supported class is enough.

3 The exact shifted-zeta evolution

3.1 The transfer and its causal generator

Define

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad H(p) = \xi(\tfrac{1}{2} + p). \quad (3.1)$$

For $0 < \omega \leq 1/2$, set

$$K_\omega(p) = \frac{H(p-\omega)}{H(p+\omega)}, \quad a_\omega(p) = \frac{H'(p-\omega)}{H(p-\omega)} + \frac{H'(p+\omega)}{H(p+\omega)}. \quad (3.2)$$

Logarithmic differentiation gives $\partial_\omega K_\omega = -a_\omega K_\omega$. The functional equation gives $|K_\omega(i\tau)| = 1$ wherever the ratio is defined. This boundary identity alone does not give a bounded causal transfer on the unweighted whole line.

Instead take inverse Laplace transforms on $\text{Re } p = \sigma > 1$, where the denominator is nonzero and the Dirichlet series converge. If k_ω is the causal kernel, define

$$V_{\omega,L}f = (k_\omega * F)|_{I_L}, \quad V_{0,L} = I. \quad (3.3)$$

For positive shift the kernel is locally integrable and the finite-window operator is bounded. The exact Beta-kernel, integer comb and rational correction are given in [1, Section S1.4]; positivity of the Beta/combin part does not imply contraction of the full transfer.

Let $G_{\omega,L}$ be the compressed causal realization of a_ω on the same Laplace line. Its symmetric quadratic form is

$$Q_{\omega,L}[g] = \text{Re} \langle g, G_{\omega,L}g \rangle. \quad (3.4)$$

This notation is interpreted as a form when the operator pairing is not separately defined. The natural form domain has logarithmic Fourier weight as in Section 2; it need not be the full operator domain of the nonsymmetric generator. Smooth compactly supported tests and their positive-shift evolutions suffice for the identities below.

3.2 The equation to be realized

Causality permits the scalar transfer identity to be compressed to the finite interval: an intermediate value beyond its right endpoint cannot return to the interval, and the input has no values before its left endpoint. The exact equation is

$$\boxed{\partial_\omega V_{\omega,L} = -G_{\omega,L}V_{\omega,L}, \quad V_{0,L} = I.} \quad (3.5)$$

It is the boxed equation adjacent to the energy derivative in the shared background [16]. It holds on admissible inputs, with a strong initial limit and a right derivative at zero on smooth tests. No operator-norm derivative at zero is asserted.

The zero-shift symbol shows the matching problem explicitly:

$$a_0(p) = w_0 + 2 \int_0^\infty n_\Gamma(u)(1 - e^{-pu}) du + \frac{2}{p + 1/2} + \frac{2}{p - 1/2} - 2 \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} e^{-p \log n}. \quad (3.6)$$

Its real quadratic form is exactly $Q_{0,L}$, including the fixed local constant, the pole response and the prime delays. At positive shift the prime term in a_ω is

$$-2 \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \cosh(\omega \log n) e^{-p \log n}.$$

The transfer itself contains the fuller multiplicative integer comb, not just the prime-power part of its logarithmic derivative.

Thus (3.5) supplies a detailed prescription. A physical construction must reproduce every component of (3.6), and its finite-shift continuation, with these coefficients. Choosing a contour with a suggestive gamma spectrum addresses only one component.

3.3 The accumulated defect and the central limit

Put

$$D_{\omega,L} = I - V_{\omega,L}^* V_{\omega,L}. \quad (3.7)$$

The product rule at positive shift yields the second evolution equation

$$\partial_\omega D_{\omega,L} = V_{\omega,L}^* (G_{\omega,L}^* + G_{\omega,L}) V_{\omega,L}, \quad D_{0,L} = 0. \quad (3.8)$$

The right side is the pullback of the symmetric form $2Q_{\omega,L}$. Equivalently one uses the bounded positive-shift product-rule expression $(G_{\omega,L} V_{\omega,L})^* V_{\omega,L} + V_{\omega,L}^* (G_{\omega,L} V_{\omega,L})$. This avoids assuming that the separate adjoint generator acts on every evolved input.

For $f \in C_c^\infty(I_L)$, integration and the initial limit give

$$\langle f, D_{\omega,L} f \rangle = \|f\|^2 - \|V_{\omega,L} f\|^2 = 2 \int_0^\omega Q_{s,L} [V_{s,L} f] ds, \quad (3.9)$$

$$Q_{0,L}[f] = \lim_{\omega \downarrow 0} \frac{\langle f, D_{\omega,L} f \rangle}{2\omega}. \quad (3.10)$$

The running shift s changes both the form and the input to it. The analytic justification in the shared background uses the gamma-ratio bounds $|\partial_\omega^j K_\omega(\sigma + i\tau)| \leq C_\sigma (1 + \log(2 + |\tau|))^j (2 + |\tau|)^{-\omega}$ for $j = 0, 1, 2$: these give positive-shift smoothing and differentiability, and dominated convergence on smooth inputs at zero. No inverse of $V_{\omega,L}$ is needed.

The positivity target is $D_{\omega,L} \succeq 0$, or equivalently $\|V_{\omega,L}\| \leq 1$. Requiring all the instantaneous $Q_{s,L}$ to be positive would suffice, but is a stronger demand than positivity of their accumulated pullback in (3.9). This distinction determines the proposed role of the reflected network in Section 8.

4 The Wilson–Loewner realization hypothesis

4.1 The later arithmetic matching problem

The fixed-theory question starts with a driver, its trace and a specified Wilson observable. Section 5 derives its first quantum equations. If an appropriate boundary construction is found, the following stronger matching requirement can then be tested.

A physical candidate must specify boundary input and output spaces, a field theory and regulator, endpoint states, a reference color sector, a physical adjoint, a family or ensemble of contours, and a normalization. After a fixed identification of its boundary spaces with $L^2(I_L)$, denote its averaged transfer by $\mathcal{V}_{\omega,L}$. It must act linearly on arbitrary boundary inputs, not merely return one scalar expectation for a single curve.

Research hypothesis 4.1 (Prescribed effective evolution). There is an independently specified Wilson boundary construction, with contours generated or parametrized by a Loewner evolution, whose normalized transfer is strongly initialized by $\mathcal{V}_{0,L} = I$ and satisfies

$$\partial_\omega \mathcal{V}_{\omega,L} = -G_{\omega,L} \mathcal{V}_{\omega,L} \quad (4.1)$$

on the arithmetic smooth core, in a causal evolution class for which this initial-value problem is unique.

Under these conditions $\mathcal{V}_{\omega,L} = V_{\omega,L}$. The existence of the construction, closure into the specified $G_{\omega,L}$, and the applicable uniqueness statement are all obligations of the hypothesis. Declaring $-G$ to be an abstract connection gives a formal ordered transport, but supplies none of the field or endpoint data. At fixed L the causal convolution generators formally commute on a common admissible core; the nontrivial question remains how a color-ordered field observable reduces to this boundary evolution.

For this later matching problem, the arithmetic differential equation is intended to *select* admissible physical and geometric data. There may be no suitable scalar driver within a proposed family, or several such families. An explicit real driver or a driver law would be an output of solving that matching problem. We do not assert that the arithmetic equation already defines one.

4.2 The insertion identity that must close

For smooth matrix fields on a regular curve, set

$$\mathcal{M} = iA_\mu \dot{X}^\mu + \Phi v, \quad \Phi = n^I X_I, \quad v = |\dot{X}|.$$

Ordered Wilson transport obeys $\partial_s U(s, 0) = \mathcal{M}(s) U(s, 0)$. For a shape change $\eta = \delta X$ and a change δn , its exact variation is

$$\begin{aligned} \delta U(1, 0) &= C_1 U(1, 0) - U(1, 0) C_0 + \int_0^1 U(1, s) \mathcal{I}(s) U(s, 0) ds, \\ \mathcal{I} &= iF_{\nu\mu} \eta^\nu \dot{X}^\mu + v D_\nu \Phi \eta^\nu + \Phi \frac{\dot{X} \cdot \dot{\eta}}{v} + v X_I \delta n^I, \quad C_s = iA_\nu \eta^\nu. \end{aligned} \quad (4.2)$$

The boundary terms combine with the variations of actual endpoint fields and their polarizations. The proof, conventions, tangent-map version and endpoint formula are in [1, Section S4].

For a fixed normalized quantum measure, $\partial_t \langle \mathcal{O}_t \rangle = \langle \partial_t \mathcal{O}_t \rangle$ whenever differentiation is justified. Holding a field configuration fixed while deriving (4.2) is the intermediate functional calculation, not a replacement of the quantum average. If an action or an external normalization is varied, its connected insertion or normalization derivative must also be included.

To obtain (4.1), the averaged insertions and all normalization terms must act on boundary inputs as $-G_{\omega,L} \mathcal{V}_{\omega,L}$. Field equations, Ward identities or a controlled reduction must establish this closure. A universal formula depending only on the curve cannot do so: in the abelian example $A_x = cy$, a straight line at $y = 0$ has $U = 1$ for every c , but its transverse variation depends on c . This obstruction concerns bare transport; it leaves open a field-dependent averaged closure.

4.3 Three parameters that must be related explicitly

Use the chordal equation

$$\partial_t g_t(z) = \frac{2}{g_t(z) - u_t}, \quad \text{hcap} = 2t, \quad (4.3)$$

where u_t is a real driver, distinguished from Wilson transport U . For a smooth reference curve $\zeta(s)$ in a region where $f_t = g_t^{-1}$ is regular,

$$X_t(s) = f_t(\zeta(s)), \quad \partial_t X_t(s) = -\frac{2f'_t(\zeta(s))}{\zeta(s) - u_t}.$$

These velocities can be inserted into (4.2); they do not define transport at a singular growing tip. A stochastic trace would need additional regularization and a defined limiting observable.

At fixed L , a dictionary $\omega = \omega(t)$ would require

$$\partial_t \mathcal{V}_{\omega(t),L} = -\dot{\omega}(t) G_{\omega(t),L} \mathcal{V}_{\omega(t),L}. \quad (4.4)$$

Loewner capacity t , arithmetic shift ω , and support length L have different meanings. Even the explicit constant-driver semicircle dictionary in [1, Section S3] identifies neither t nor the logarithmic endpoint radius with ω . When L changes, so do the space and compression; the fixed-window equation (4.4) omits that change. Section 9 treats it as a separate finite gluing step.

5 Quantum evolution along a genuine growing trace

The first task in the physical program is to determine the quantum evolution induced by a specified Loewner curve within a fixed theory. This does not require an explicit expectation value or an immediate identification with the arithmetic transfer. We now carry out that calculation for an independently specified observable in pure four-dimensional Yang–Mills theory. Unlike the regular inverse-map deformations in Section 4, the path here is the actual growing trace; previously traced points remain fixed.

5.1 A fixed theory and a specified tip contraction

Fix Euclidean $SU(N)$ Yang–Mills at $\theta = 0$, including its coupling and state. Embed the upper half-plane in $x_3 = x_4 = 0$ and let a deterministic driver $u(t)$ generate a sufficiently regular trace $q[0, t]$, starting at $q(0) = 0$. The slit is geometric data, not a moving physical boundary of the quantum theory. The driver need not be random for the field expectation to be quantum.

An open holonomy carries color indices at its endpoints. We contract them by returning from $q(t)$ to the origin along the straight chord. This return path is part of the observable. Let U_t transport along the actual prefix and $R_t(r)$ transport along the straight segment from 0 to $rq(t)$. Define

$$\mathbf{H}_t = R_t(1)^{-1} U_t, \quad W_\tau(t) = \frac{1}{N} \langle \text{tr } \mathbf{H}_t \rangle, \quad W_\tau(0) = 1. \quad (5.1)$$

To state smooth-field identities with a controlled short-time expansion, the connection is the Yang–Mills flowed field B_τ at a fixed positive resolution τ . The action and measure remain those of the original interacting theory; there is no Gaussian substitution. Field-flow time τ is distinct from Loewner time t and the arithmetic shift ω [31, 32].

The following identities are exact for smooth connections. Their expectation-value versions assume existence of the selected regulated or flowed correlators and the integrability needed to

differentiate them. No nonperturbative continuum construction is asserted. Removing the resolution requires a separate treatment of cusp and contact terms. Neither this observable nor its normalization is identified with $V_{\omega,L}$.

5.2 The tip cancellation and the quantum hierarchy

Suppress τ in field notation and transport the curvature back to the fixed basepoint:

$$\hat{F}_t(r) = R_t(r)^{-1} F_{12}(rq(t)) R_t(r), \quad J_t = \int_0^1 r \hat{F}_t(r) dr, \quad k_t = q_1 \dot{q}_2 - q_2 \dot{q}_1. \quad (5.2)$$

Prefix growth inserts the connection at the tip. Differentiating the return chord produces the opposite endpoint term, so these terms cancel before averaging. The remaining exact matrix and expectation equations are

$$\boxed{\dot{H}_t = ik_t J_t H_t, \quad \dot{W}_\tau(t) = ik_t M_1(t), \quad M_1 = \frac{1}{N} \langle \text{tr}(J_t H_t) \rangle.} \quad (5.3)$$

The geometric coefficient is twice the oriented area growth of the trace with its chord. The quantum coefficient is a curvature insertion in the same interacting expectation.

Differentiating once more gives

$$\dot{M}_1 = \dot{q}^\mu M_{D,\mu} + 2ik_t M_2, \quad \mu = 1, 2. \quad (5.4)$$

Here $M_{D,\mu}$ is the normalized expectation with insertion $\int_0^1 r^2 \widehat{D}_\mu F_{12_t}(r) dr$; M_2 has the ordered insertion

$$\int_{0 < s < r < 1} r s \hat{F}_t(s) \hat{F}_t(r) ds dr.$$

Both are followed by H_t inside the trace. The order and the factor of two follow from the changing radial transports. The complete derivation and domains are given in [1, Section S13]. These are equations for quantum expectations without a Wick rule or large- N factorization; they are not yet a closed scalar equation for W_τ .

5.3 A linear driver and a controlled short-time response

For the concrete driver $u(t) = at$, the upper-half-plane tip obeys

$$aq + 2 \log(1 - aq/2) = a^2 t, \quad q(t) = 2i\sqrt{t} + \frac{2a}{3}t - \frac{ia^2}{18}t^{3/2} + O_a(t^2). \quad (5.5)$$

The implicit relation is for $a \neq 0$ on the branch issuing from the origin; its $a = 0$ limit is the vertical slit. It follows by integrating the forward Loewner equation, not by evaluating that equation at its singular tip. The enclosed oriented area is $\mathcal{A}(t) = -(2a/9)t^{3/2} + O_a(t^{5/2})$. Under the local moment assumptions stated in the companion,

$$\boxed{W_\tau(t) = 1 - \frac{2a^2}{81} C_\tau t^3 + O_{\tau,a}(t^{7/2}), \quad C_\tau = \frac{1}{N} \langle \text{tr} F_{12}[B_\tau](0)^2 \rangle.} \quad (5.6)$$

The coefficient C_τ is a gauge-invariant moment in the fixed theory. Its value need not be known to derive this relation. The companion bounds the remainder explicitly by curvature and covariant-derivative moments; the expansion is at fixed τ . For the constant driver the chosen loop backtracks exactly, so $H_t = I$ and $W_\tau(t) = 1$ configuration by configuration. That control concerns this return-path choice, not every Wilson observable on a straight trace.

5.4 The Makeenko–Migdal closure question

Schwinger–Dyson identities relate field-equation insertions to derivatives of observables. Their loop-equation form is a natural candidate for reducing (5.4) [34]. The first missing terms are explicit. In four dimensions $D_1 F_{12}$ is only one part of $E_2 = \sum_\mu D_\mu F_{\mu 2}$: the terms $D_3 F_{32}$ and $D_4 F_{42}$ remain even though the contour is planar. Finite- N splitting terms also retain expectations of products. At positive τ , differentiating the observable with respect to the original field differentiates the flow map $B_\tau[A]$ as well. These terms have not been reduced to W_τ and M_1 .

Thus the calculation supplies a concrete relation between genuine Loewner growth and quantum evolution, and identifies the correlators needed for a closure attempt. It supplies neither a closed Makeenko–Migdal reduction nor the physical arithmetic transfer. The freedom to choose another return path, endpoint contraction or insertion stays within the fixed-theory question; each choice must have its own derived evolution.

6 A fixed-window test and the localization question

The first realization test can be made before the first arithmetic delay. For $0 < L < \log 2$, no nontrivial integer translation acts on I_L . The local term and both rational pole terms remain. On this window we can state the complete target, calculate a normalized Wilson readout, and exclude that readout within a precise regularity class. We can also formulate a localization test for the required cancellations. The detailed calculations and proofs are in [1, Sections S10–S12].

6.1 The local and pole terms constrain the response

Let $A_L = -G_{0,L}$ be the required first shift derivative. Away from the diagonal its causal kernel is

$$b(u) = 2 \frac{e^{-u/2}}{1 - e^{-2u}} - 4 \cosh(u/2) = 2 \frac{e^{-5u/2}}{1 - e^{-2u}} - 2e^{u/2}, \quad 0 < u < L. \quad (6.1)$$

The first pole cancels the lowest gamma mode. The other leaves the signed growing exponential in the second expression. The kernel behaves as $1/u$ at zero; it must be interpreted together with a contact subtraction. In the hard-delay-cutoff convention, on $C_c^\infty(I_L)$,

$$(A_L f)(x) = \lim_{\epsilon \downarrow 0} \left[c_\epsilon f(x) + \int_\epsilon^{x+L/2} b(u) f(x-u) du \right], \quad c_\epsilon = \log \epsilon + \gamma + \log(2\pi) + O(\epsilon). \quad (6.2)$$

An integral whose upper limit is at most ϵ means zero. This finite part is fixed by w_0 in the stated cutoff scheme, not a new normalization choice. Its derivation, including the zero-extension tail, is in [1, Section S10.1].

At positive shift, absorb the first rational factor into the gamma ratio. The factor whose compression is the full arithmetic transfer on this window becomes

$$K_\omega^<(p) = \pi^\omega \frac{\Gamma((p+5/2-\omega)/2) p-1/2-\omega}{\Gamma((p+5/2+\omega)/2) p-1/2+\omega}. \quad (6.3)$$

This is an arithmetic identity, not yet a field-theory construction. Its locally integrable positive-shift kernel has a singularly concentrating identity limit; there is no separate delta channel at positive shift [1, Section S10.2].

6.2 A normalized causal Wilson readout that fails the test

Use the ordinary reflected endpoint pairing, with the physical polarizations $q_2, \bar{q}_2$, a common reference color fiber and the adjoint scalar map $N = -MR$. For the smooth polynomial bulges of the companion, deform a positive height by $\eta(\omega) = \eta_0 + \omega$, $\eta_0 > 0$, keeping the endpoints fixed. Let $K_\omega(x, y)$ be the radially weighted reflected correlator. A definite linear response is

$$(\mathcal{V}_{\omega,L}f)(x) = f(x) + \int_{-L/2}^x [K_\omega(x, y) - K_0(x, y)]f(y) dy. \quad (6.4)$$

It has exact identity initial data and causal support. The retarded integration in logarithmic radius is part of the specified readout, not a consequence of Euclidean reflection. It avoids inverting the compact free Gram operator.

In the independent abelian Gaussian probe model defined in [1, Sections S10.3–S10.4], the free endpoint term has zero variation, while the Wilson dressing gives a nonzero variation at positive bulge height. The unequal-radius calculation and a uniform bound show that $\partial_\omega \mathcal{V}_{0,L}$ is nevertheless a bounded integral operator. It therefore cannot equal $-G_{0,L}$ on the smooth core: for $f_N(x) = e^{iNx}\chi(x)$ with nonzero $\chi \in C_c^\infty(I_L)$,

$$\frac{Q_{0,L}[f_N]}{\|\chi\|^2} = \log N - \log(2\pi) + o(1).$$

No bounded derivative, even with a bounded local multiplier, has this quadratic-form growth. At positive shift there is a second obstruction: (6.4) is identity plus a compact operator, whereas $V_{\omega,L}$ is compact. The proofs and regulator scope are in [1, Section S10.5].

These exclusions do not cover all Wilson constructions or a controlled singular limit of regulated responses. They show what must change in the tested observable-to-operator map. Localizing the same regular readout would not remove the obstruction.

6.3 Localization as a test of the surviving modes

Localization is a candidate mechanism for deriving the cancellations in (6.1). Pestun's fluctuation calculation organizes cancellations by an equivariant complex [22]. Baker's open-line discussion already points toward a lower-dimensional defect theory [6, Section 3.3]. Wang's defect-localization framework contains endpoint bilocals joined by gauge transport [23, Section 5.1, equation (5.5)]; its unevaluated normal-fluctuation determinant remains an explicit assumption [23, Section 4.4].

An auxiliary calculation makes the proposed test concrete. Put $a_n = 2n + 1/2$ and retain one complex bosonic Gaussian for each $n = 1, \dots, N$, together with one complex Grassmann pair:

$$Z_N(p) = (p - 1/2) \prod_{n=1}^N (p + a_n)^{-1}. \quad (6.5)$$

Then exact finite-product algebra gives

$$\pi^\omega \lim_{N \rightarrow \infty} N^{-\omega} \frac{Z_N(p - \omega)}{Z_N(p + \omega)} = K_\omega^<(p). \quad (6.6)$$

Removing the lowest boson accounts for the first pole cancellation, and the Grassmann factor supplies the remaining rational ratio. These modes and the normalization have been chosen from the arithmetic target. Their presence is not a localization theorem or even a specified supersymmetric action.

The physical test is whether a defect-compatible supercharge, boundary conditions and observable produce these surviving contributions and the required finite normalization. An especially sharp comparison, independent of an additive p -independent generator constant, is

$$\partial_p a_0^<(p) = 2 \sum_{n=1}^{\infty} \frac{1}{(p+2n+1/2)^2} - \frac{2}{(p-1/2)^2}. \quad (6.7)$$

The construction and its missing physical dictionary are spelled out in [1, Section S11]. A localization deformation coefficient multiplying a Q -exact term cannot simply be identified with the physical shift. Likewise a partition ratio does not yet act on arbitrary boundary inputs. The original conformal semicircle remains a relevant control, outside the restricted constant-Poincare-charge endpoint obstruction.

6.4 A predictive boundary calculation and its spatial limitation

A specified free three-dimensional $\mathcal{N} = 4$ hypermultiplet on a hemisphere provides a concrete localization control [25, 26]. With its Higgs boundary polarization, put $\epsilon = (4\pi r)^{-1}$ and $x = |Y|^2/\epsilon$. The vacuum is proportional to e^{-x} , whose boundary Fourier–Mellin amplitude is $\Gamma(s)/\sqrt{2\pi}$ at $s = 1/2 - i\sigma$. Separated scalar endpoints with identity transport give $\epsilon x e^{-x}$ and hence $\epsilon \Gamma(s+1)/\sqrt{2\pi}$. The gamma factor and this insertion follow from the selected physical problem; they do not require the arithmetic tower to be prescribed. Their derivation, ordering and ordinary-adjoint checks are in [1, Sections S12.1–S12.2].

The transform acts on a boundary field’s magnitude. It does not yet transform spatial logarithmic radius. In an adapted flat frame, a protected scalar is $O(t) = q_1(t) + tq_2(t)/(2r)$, with $\mathcal{Q}q_2(t) = \chi(t)$ a nonzero free fermion. Ordinary spatial dilation gives

$$\mathcal{Q}[D, O(t)] = -\frac{t}{2r} \chi(t). \quad (6.8)$$

Thus the fixed protected representation does not receive the ordinary radial evolution on arbitrary closed representatives. The compensating R rotation restores closure but gives the exact twisted dilation, trivial on the tested local classes. The centered hemisphere geometry and the scope of this obstruction are in [1, Section S12.3].

There is a distinct canonical-Hodge alternative. Selecting the kernel of $D - R_H$ first permits a restricted radial semigroup. For an elementary scalar, however, a spatial harmonic has $D = \ell + 1/2$ and $R_H = 1/2$, so only $\ell = 0$ survives. The projected endpoint response is $e^{-u/2}$. The omitted even-angular channel is

$$\frac{e^{-5u/2}}{1 - e^{-2u}},$$

precisely the positive tower appearing in (6.1). Composite protected states can have a matching counting character after additional selections, but an elementary endpoint does not excite that character. A trace, a boundary amplitude and an ordinary-adjoint response are different tests [1, Sections S12.4–S12.5].

These calculations provide a physical precedent for a gamma amplitude and a specific obstruction to its spatial interpretation in this free model. They do not rule out an enlarged or interacting sector. Any continuation must specify how it retains the missing modes and their endpoint matrix elements, before fitting a spectral-parameter map or finite normalization. A polynomial-degree comparison using an unproved affine map is conditional; changing the offset can change that degree [1, Section S12.6].

Finally, signed mode cancellation does not establish an ordinary positive norm. The independent cumulative-storage identity of Section 8 and the spatial gluing estimates of Section 9 remain separate requirements. Neither this test nor the auxiliary product supplies an arithmetic Loewner driver.

7 What the existing calculations contribute

The previous calculations provide candidate ingredients and restrictions on a realization. They do not yet supply the effective transfer of Hypothesis 4.1.

7.1 A spectral ingredient, with a fixed subtraction sign

A free three-dimensional defect scalar in logarithmic radial coordinates has angular rates $\ell + 1/2$. Even Poisson smearing selects $\ell = 2n$ and gives, for $0 \leq \rho < 1$,

$$C_\rho(u) = \sum_{n \geq 0} \rho^{4n} e^{-a_n |u|} = \frac{e^{-|u|/2}}{1 - \rho^4 e^{-2|u|}}. \quad (7.1)$$

This is a positive covariance tending to $n_\Gamma(|u|)$ off the diagonal. If T_ρ denotes convolution by C_ρ and $M_\rho = \hat{C}_\rho(0)$, the positive difference form is $M_\rho I - T_\rho$, whose limit is $\mathcal{E}_\Gamma$. The opposite subtraction has the wrong high-frequency sign, and a finite contact constant cannot repair it. The auxiliary Gaussian mode model also produces the gamma partition ratio, but neither a physical projection to the required multiplicities nor its identification as a boundary transfer is established [1, Section S2].

These results explain why defect matter is relevant and fix a sign that a candidate must respect. They supply neither w_0 by themselves nor the pole and arithmetic terms in (3.6). The auxiliary completion in Section 6 includes both pole factors by a prescribed change of modes; their physical origin remains open. The exact positive Beta/comb factorization likewise retains an essential rational subtraction [1, Section S1.4].

7.2 Contour freedom and endpoint compatibility

The inherited common-reference semicircles become vertical slits after inversion and have constant Loewner driving. Their rigidity holds for one fixed scalar prescription and one specified bulk charge. A tangent-dependent scalar coupling changes that restriction: with the full chiral D3–D5 projector, an oriented real tangent isometry admits a common complex charge space of dimension two in a normal plane and dimension one across all defect directions. These are exact statements in the stated Euclidean Clifford module [1, Sections S3 and S5].

Contour freedom does not complete the observable. In this constant-map Poincaré-charge ansatz, ordinary scalar endpoints annihilated by the same charge have zero mutual free contraction. Their bilinear also has nonzero residual R charge, giving a vanishing expectation in an invariant vacuum. Pairing an endpoint with its physical adjoint restores a nonzero contraction but changes the supersymmetry conditions [1, Section S6]. Baker’s original conformal semicircle is a nonzero control outside that restricted ansatz, not an excluded construction.

A candidate should therefore be tested for a nonzero normalized endpoint pairing before trying to extract its arithmetic generator. A shared line supercharge alone does not ensure such a pairing.

7.3 A nontrivial response, with an unfinished physical sum

Ordinary time reflection gives a consistent classical adjoint and a positive free endpoint kernel. A smooth polynomial bulge can join its reflected branch without a fixed-angle cusp. In the direct

Feynman-gauge Gaussian bulk-exchange sector, the stripped self-response is

$$I(\eta) = \left(\frac{3}{2} - 2 \log 2\right) \eta^2 + O(\eta^4), \quad \frac{3}{2} - 2 \log 2 > 0. \quad (7.2)$$

The construction and remainder estimate are in [1, Sections S7–S8]. Endpoint-to-line exchange, defect interactions, self-energies and counterterms are still needed for the full response. Thus the coefficient is a concrete insertion test, not an effective arithmetic generator or an all-order protection theorem.

The distinction between supersymmetry and a positive norm is visible even abstractly: $Q\psi = 0$ and $\delta\psi = Q\lambda$ give $\delta\|\psi\|^2 = 2 \operatorname{Re} \langle Q^\dagger \psi, \lambda \rangle$, which need not vanish. A successful reduction must handle both the chosen charge and the physical adjoint.

8 Reflection as a possible explanation of cumulative positivity

8.1 The identity the positive pairing must supply

The second construction target is a positive Hilbert space $\mathcal{H}_{\omega,L}$ and a linear amplitude map $\mathcal{B}_{\omega,L} : L^2(I_L) \rightarrow \mathcal{H}_{\omega,L}$ such that

$$\boxed{\|f\|_{L^2(I_L)}^2 = \|\mathcal{V}_{\omega,L} f\|_{L^2(I_L)}^2 + \|\mathcal{B}_{\omega,L} f\|_{\mathcal{H}_{\omega,L}}^2.} \quad (8.1)$$

This is a proposed physical balance law. Once the transfer matching has been proved, it implies

$$D_{\omega,L} = \mathcal{B}_{\omega,L}^* \mathcal{B}_{\omega,L} \succeq 0. \quad (8.2)$$

A proof first on a dense core, with bounded extensions, is enough. The initial normalization forces $\mathcal{B}_{0,L} = 0$ if the identity extends to zero shift.

The input and output norms in (8.1) are the fixed, unweighted arithmetic norms. A parameter-dependent physical metric requires an explicit isometric identification and the extra terms caused by its variation. Defining $\mathcal{B} = D^{1/2}$ after assuming $D \succeq 0$ would restate the desired conclusion; the amplitudes must be specified independently. A whole-line unitary model available only under RH also cannot be the starting proof.

8.2 What ordinary reflection currently establishes

For tangent map M and time reflection $R = \operatorname{diag}(-1, 1, 1, 1)$, the adjoint branch uses $N = -MR$ and reversed path ordering. If F_i is the Wilson-dressed endpoint vector at a common reference, the proposed network is

$$K_{ij} = \langle \Theta_0 F_i F_j \rangle.$$

A regulator with the appropriate reference color sector and Osterwalder–Schrader inequality would make this a Gram matrix. The interacting regulator, operator domain and positivity-preserving renormalization are assumptions, not established consequences of gauge invariance. The full construction and its scope are in [1, Section S7].

At free endpoint order the time-ray kernel is unconditionally positive:

$$K^{(0)}(r, s) = \frac{1}{r + s} = \int_0^\infty e^{-\lambda r} e^{-\lambda s} d\lambda, \quad r, s > 0. \quad (8.3)$$

After radial weights it gives $G_o(u) = 1/(2 \cosh(u/2))$. This is only the opposite-ray part of $n_\Gamma(|u|) = (G_s(u) + G_o(u))/2$, where $G_s(u) = 1/(2 \sinh(|u|/2))$. Moreover, a positive kernel of states

has not been identified with the defect $D_{\omega,L}$. It needs a specified shift-dependent amplitude and the exact balance (8.1). The unmodified free kernel does not satisfy the required vanishing at zero shift.

Internal rotations that align the two scalar maps need not preserve this positive metric; explicit endpoint-doublet and odd-H counterexamples are retained in the companion. The positive ordinary adjoint is therefore a substantive part of the proposal.

8.3 Why the target is the accumulated norm

Equation (3.9) integrates the energy on the moving input $V_{s,L}f$. A reflected realization might explain the resulting positive norm without factoring each instantaneous $Q_{s,L}$. The shifted-zeta storage calculation records a useful distinction: at $L = \log 7$ and $\omega = 10^{-11}$, one exact polynomial input has a negative instantaneous quotient of about -2.998×10^{-27} but a positive cumulative quotient of about 8.7170×10^{-38} [20, Appendix A]. These inherited enclosures concern one input, not an all-input contraction bound. Nor do they show negativity of the actual evolving integrand, since $Q_{\omega,L}[f]$ and $Q_{\omega,L}[V_{\omega,L}f]$ use different inputs.

The example motivates allowing accumulated storage to carry the positivity argument. It does not establish the physical balance law or justify reversing a forward dissipation inequality.

9 The depth–shift continuation problem

9.1 A sufficient diagonal route

The end of the program requires a sequence of operator bounds, not a particular smooth curve in the parameter plane.

Proposition 9.1 (Sufficient depth–shift route). *Suppose $L_j \rightarrow \infty$, $0 < \omega_j \downarrow 0$, and $\|V_{\omega_j,L_j}\| \leq 1$ for every j . Then $Q_{0,L}[f] \geq 0$ for every $L > 0$ and $f \in C_c^\infty(I_L)$, and hence RH holds.*

Proof. Fix L and f . For large j , zero-extend f to I_{L_j} . Restriction of its output to I_L is $V_{\omega_j,L}f$, because the causal kernel is the same. Restriction cannot increase its norm, so $\|V_{\omega_j,L}f\| \leq \|f\|$. Divide its nonnegative defect by $2\omega_j$ and take the fixed-test limit (3.10). Theorem 2.1 completes the implication. $\square$

No rate coupling L_j to ω_j , uniform positive margin, or differentiable path is needed for this implication. Every bound must hold on all inputs at its selected pair. Finite matrices alone do not supply that assertion.

9.2 What is meant by a critical path

Here a critical path means a continuation along which a construction or rigorous estimate remains effective as depth grows and shift decreases. It is not an assumed boundary of the true contraction region. Under RH the positive-shift quotient is inner, so finite-window contraction holds throughout the parameter range; the parent Wilson-lines discussion makes this distinction [12]. There is also an unconditional anchor at $\omega = 1/2$ from the known innerness of the quotient [21, Section 1.4]. This anchor gives finite-window contraction, but does not automatically initialize the strict margins or physical data needed for a chosen continuation.

The earlier adaptive-shift estimate is a different, sufficient route. Assuming $Q_{0,L} \succeq m_L I > 0$, and writing $X_L f = x f$ on the centered interval, put

$$\kappa_L = \sup_{f \neq 0} \left(\frac{Q_{0,L}[X_L f]}{Q_{0,L}[f]} \right)^{1/2}. \quad (9.1)$$

With a finite proved bound on this form-multiplication norm, the inherited estimate is

$$Q_{\omega,L} \succeq \cos(2\omega\kappa_L) Q_{0,L}, \quad 0 \leq \omega\kappa_L \leq \pi/4 \quad [18]. \quad (9.2)$$

It gives a small-shift schedule after central positivity has been proved at that depth. It does not establish the central positivity needed to enlarge the interval. A lower estimate of κ_L from finitely many vectors cannot justify that schedule.

9.3 Length extension has its own exact constraint

Translate the interval to $(0, L+h)$ and split it into old and new pieces. Causality gives

$$V_{\omega,L+h} = \begin{pmatrix} X & 0 \\ Y & Z \end{pmatrix}, \quad X = V_{\omega,L}, \quad Z = V_{\omega,h}. \quad (9.3)$$

If X and Z are strict contractions in operator norm, set

$$E = I - X^* X, \quad F_{\text{out}} = I - Z Z^*.$$

The inherited cumulative-storage criterion is

$$\boxed{\|V_{\omega,L+h}\| \leq 1 \iff \|F_{\text{out}}^{-1/2} Y E^{-1/2}\| \leq 1.} \quad (9.4)$$

It measures cross-interval transfer in the old-input and new-output defect metrics. A strict inequality gives a strict contraction. The proof and the simultaneous depth-shift criterion are in [1, Section S9], following [19]. The diagonal strictness is required for the displayed inverses; a semidefinite boundary case needs a separate range formulation.

This suggests a precise physical gluing target: derive or bound the normalized cross map in (9.4), preserving its directions and cancellations. Separate worst-case scalar estimates can lose the information a continuation needs. If the shift also changes, the old and new defect metrics change as well; the finite-step Schur test in the companion keeps those changes. The fixed- L shift ODE alone is not that joint step.

For reflection-compatible coordinates one can use

$$P_{\omega,L} = \frac{2}{\omega} \operatorname{Re}[(I - V_{\omega,L})(I + V_{\omega,L})^{-1}].$$

Here $\operatorname{Re} A = (A + A^*)/2$ and the inverse exists for the positive-shift finite-window Volterra operator. The exact congruence

$$D_{\omega,L} = \frac{\omega}{2} (I + V_{\omega,L}^*) P_{\omega,L} (I + V_{\omega,L}) \quad (9.5)$$

preserves positivity and the normalized spatial coupling. These are useful coordinates, not an additional positivity result. At positive shift P is bounded, whereas $Q_{0,L}$ is an unbounded form; a core expansion does not give a uniform operator-norm approximation.

9.4 Separate fixed-window arithmetic progress

The separate EMA calculation now supplies an internally certified all-input bound at a designated small window:

$$Q_{0,1/2} \succeq I/40, \quad D_{10^{-3},1/2} \succeq 0.000049998 I. \quad (9.6)$$

It uses the fixed arithmetic tower, exact endpoint response and rigorous bounds on omitted inputs, with rational interval checks at two precisions [27]. Specialist review remains open. This closes the pilot's input-complement gap, without claiming a new depth horizon or an all-depth result.

For the adjoining window $h = 1/20$ at the same shift, the diagonal floor transfers to both defect orientations. The scalar sufficient estimate loses a factor between 55 and 72, and the finite quotient near 0.802 is not an all-input upper bound [28]. A subsequent energy-weighted operator calculation establishes internally, at this designated append,

$$\left\| F_{\text{out}}^{-1/2} Y E^{-1/2} \right\| < 0.951 < 1 \quad [29].$$

Its forward/backward energy argument controls the entire input complements; specialist review remains outstanding. It closes this bounded append test, without a new positivity horizon or an all-depth continuation. Its proof remains in the separate critical-path investigation. Neither it nor the new physical hierarchy in Section 5 identifies a Wilson expectation with the arithmetic transfer.

9.5 The all-depth requirement

A recursion with $L_{j+1} = L_j + h_j$ must establish $\sum_j h_j = \infty$ as well as $\omega_j \downarrow 0$. Local extendibility could otherwise accumulate at a finite length. Arithmetic delays and crossings of weak directions favor a finite step criterion over an assumed smooth critical-curve ODE. Decreasing the shift also cannot be justified by reversing forward dissipation; no bounded inverse of $V_{\omega,L}$ is assumed.

Numerical exploration should retain complete output norms and a rigorous bound on the unresolved input complement. In particular, $I - (\text{IVII})^*(\text{IVII})$ can look positive because output was discarded. The parent Loewner assembly is a reusable diagnostic tool, but a finite experiment neither proves (9.4) on the full space nor prevents finite-depth accumulation.

10 Research agenda and present status

The current physical result is a quantum insertion hierarchy for a genuine Loewner trace in one fixed interacting theory. The return chord specifies its color contraction, the moving-tip terms cancel, and a linear driver has a short-time response with an explicit smooth-field remainder bound. The quantum identities assume the stated regularity and moments at fixed field resolution. They do not yet close on the scalar expectation.

10.1 The next discriminating tasks

The immediate task is to apply finite-regulator Schwinger–Dyson equations to the derivative moment in (5.4). Retain the transverse field components, finite- N product expectations and, at positive field-flow time, the derivative of $B_\tau[A]$. The concrete question is whether these terms form a useful closed observable family, or leave a residual that can be bounded. Deriving that equation does not require first computing the expectation value explicitly. A change of return path, endpoint contraction or insertion should be assessed within the fixed theory before proposing a different bulk action.

The unflowed limit is a separate task. The present short-time theorem holds at fixed positive resolution and cannot be extrapolated past uncontrolled cusp or contact terms. Likewise an $\mathcal{N} = 4$ extension must include its scalar couplings; the pure-YM backtracking control is not automatically preserved. Loop equations organize this question but do not identify an arbitrary planar Loewner map with a symmetry of the four-dimensional theory.

The earlier free and Gaussian tests remain scoped controls. The regular readout fails the arithmetic singularity and compactness comparison; the free localized hemisphere fails direct spatial descent, and its canonical Hodge projection loses higher elementary modes. Those results do not exclude other observables in an interacting theory. A counting character, a scalar expectation and an operator on arbitrary boundary inputs still require distinct comparisons.

If a useful quantum evolution closes, a later boundary construction can be tested against the complete arithmetic generator and its fixed normalization. A separate positive-norm identity would still be required. The arithmetic anchor and bounded append certificate are contextual results in the critical-path investigation [27, 29]; they do not provide that physical identification or an all-depth theorem. The new hierarchy makes progress on the physical relation without making an arithmetic match an entrance condition for every calculation.

10.2 A compact statement ledger

Ingredient	Present status
Arithmetic shift and defect ODEs	Exact inherited identities, with their causal and form domains.
Weil limit and diagonal implication	Background positivity criterion and a sufficient conditional route to RH; the required unbounded family of contractions remains open.
Free tower and gamma model	Explicit spectral ingredients with a fixed subtraction sign; no complete transfer or physical mode selection.
Contour and Wilson variation tools	Exact smooth identities and scoped rigidity restrictions; genuine prefix growth is treated separately from inverse-map deformation.
Genuine-trace quantum hierarchy	Exact smooth-field tip cancellation and first two insertion equations; quantum averages require stated regularity. Scalar closure remains open.
Linear-driver short-time response	Explicit geometric coefficient and remainder bound at fixed field resolution; curvature moment unevaluated and unflowed limit open.
Chiral projectors and scalar endpoints	Compatible charges exist; the ordinary same-charge scalar pairing vanishes in the stated ansatz.
Ordinary reflected pairing	A nonzero positive free kernel; interacting positivity and its identification with cumulative storage remain open.
Smooth bulge response	A nonzero direct Gaussian bulk sector; the full interacting response and protection question remain open.
Fixed-window response test	Full local/pole comparison and two proved obstructions for the specified regular class; singular constructions remain open.
Auxiliary localization product	Exact boson–fermion product for the prime-free factor; physical mode selection and finite normalization remain open.
Predictive localization control	Gamma boundary amplitude and actual scalar-pair state; direct spatial descent fails in the specified free protected sector.
Canonical Hodge alternative	Restricted radial evolution exists; its elementary scalar channel loses the higher spatial tower. Composite traces do not supply endpoint spectral weights.
Arithmetic EMA anchor	Internally certified all-input fixed-window bounds, with two interval precisions; specialist review outstanding.
Physical transfer and positive balance	Research hypotheses, including exact generator closure, normalization, uniqueness and an independent amplitude map.
Depth–shift gluing	Exact conditional Schur criteria and one internally certified all-input append; an unbounded continuation remains open.

The companion preserves the technical evidence behind this ledger, including the original 299 finite diagnostics and the separately recorded localization and genuine-trace controls. Their scope is signs, ranks, geometry, ordered variations and arithmetic consistency. They do not evaluate the interacting path integral, verify the realization hypothesis, or certify its positive-norm identity or RH.

A Provenance and contribution disclosure

This version was restructured on 20 September 2026 by OpenAI GPT-6 (Codex), for Edward Baker, following Baker’s proposal to use the shared background’s shifted-zeta equation as the prescribed effective Wilson–Loewner evolution. It is a working research exposition, not a submitted or independently refereed paper. The author designation records for whom it was drafted and does not assert human verification of every derivation. The configured reasoning effort was not exposed to the assistant in this session and is not inferred.

The unified versions 0.1 and 0.2 remain unchanged in the dated draft archive. The explicit archimedean decomposition added in version 0.2 is retained in Section 2. Version 0.3 separates this program exposition from *Supplementary information* [1], which preserves the detailed field and contour calculations and their numerical controls. The shift-flow note [17] supplies the conceptual handoff. The shift and storage identities and the continuation algebra are inherited from [16, 18, 19], not new realization theorems of this draft.

Version 0.4 incorporates the fixed-window calculation and the localization direction proposed by Edward Baker. The new Section 6 summarizes the full local/pole comparison, the scoped regular-readout exclusions and the auxiliary boson–fermion identity proved in the companion’s Sections S10–S11. No supersymmetric action, physical mode pairing or positive norm is inferred from the auxiliary product. Both the derivations and their presentation were developed with OpenAI GPT-6 (Codex) assistance.

The literature checks and original local sources are recorded in `MANUSCRIPT_SOURCES.json`; its restructuring entry records the additional inputs at this revision. The `BUILD_RECORD.json` binds both reviewed PDFs to their sources and supporting diagnostics. Dated snapshots include both standalone documents and have their own file-identity record. These records establish provenance and reproducibility, not mathematical correctness. No independent review of either document has been completed.

Version 0.5 incorporates the predictive hemisphere calculation, the scoped radial-descent obstruction and its canonical-Hodge qualification. The one-sided gamma amplitude and Hodge theorem are external inputs [25, 26]; the spatial-channel and endpoint-response comparisons are model-assisted applications, not independent physical existence theorems. Section 9 also reports the separate EMA anchor and cumulative-append obstruction [27, 28]. The current research notes identify their checks and outstanding specialist review. The preceding versions 0.1–0.4 are preserved unchanged.

The live revision of 21 September 2026 incorporates Baker’s clarification that a specified Loewner evolution generates the curve defining the quantum observable, and that the immediate aim is its evolution equation in a fixed theory. The new chord-completed Yang–Mills calculation, its ordered hierarchy and quantitative remainder come from the research note [30]. They were derived and integrated with OpenAI GPT-6 (Codex) assistance; the reasoning effort was not exposed. The field-theory regularity assumptions, finite matrix diagnostics and outstanding specialist review are stated separately. The archived version 0.5 and all earlier snapshots remain unchanged. At Baker’s request this is an unarchived live revision: no new dated draft or version number has been saved. Arithmetic manuscript separation is deferred.

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